

1.

①. $D + 18$

②. A

③. A

④. C

⑤. D

⑥. D

2.

+28

(a) hydrogen cyanide.

(b) copper (II) ~~phosphor~~phosphate.

(c) silane. phosphate.

(d) potassium ~~acetate~~ oxalate.

(e) barium chloride dihydrate.

(f) sulfate ion.

(g) acetylene.

(h) formic acid.

(i) methylcyclopropane.

(j) m-bromochlorobenzene.

3. **tb.**

	Open	Close	Isolated
Heat	0	0	X
Matter	0	X	X

4. $W = -P_0 V$

tb. $= -nRT$

$$= -\frac{3}{65.409} \times 8.314 \times 298.15$$

$$= -113.69 \text{ J}$$

$$\Delta E = q + w$$

$$= -13980 \text{ J} + (-113.69 \text{ J})$$

$$= -14093.69 \text{ J}$$

5. $\Delta V = 0 \therefore -P_0 V = 0 = w$

tb. $\Delta E_{0^\circ\text{C}} = q = \Delta H_{\text{fus}} = 6.01 \times \frac{3}{18} = 1.002 \text{ kJ}$

$$\Delta S_{0^\circ\text{C}} = \frac{q}{T} = \frac{1002 \text{ J}}{273.15 \text{ K}} = 3.668 \text{ J/K}$$

$$\Delta E_{0^\circ\text{C} \rightarrow 25^\circ\text{C}} = 75.3 \times \frac{3}{18} \times (25 - 0) = 313.75 \text{ J}$$

$$\Delta S_{0^\circ\text{C} \rightarrow 25^\circ\text{C}} = 75.3 \times \frac{3}{18} \times (25 - 0) \div (273.15 + 25) = \frac{313.75 \text{ J}}{298.15 \text{ K}} = 1.052 \text{ J/K}$$

$\therefore$ (a) $\Delta E_{0^\circ\text{C}} = 1.002 \text{ kJ}$
 $\Delta S_{0^\circ\text{C}} = 3.668 \text{ J/K}$

(b) $\Delta E_{0^\circ\text{C}} + \Delta E_{0^\circ\text{C} \rightarrow 25^\circ\text{C}} = 1002 + 313.75 = 1315.75 \text{ J}$

$$\Delta S_{0^\circ\text{C}} + \Delta S_{0^\circ\text{C} \rightarrow 25^\circ\text{C}} = 3668 + 1.052 = 4.72 \text{ J/K}$$

6. (a) $\Delta G^\circ_{298} = -RT \ln K_p$

$\Rightarrow -141600 = -8.314 \times (273.15 + 25) \ln K_p$

$\Rightarrow \ln K_p = \frac{-141600}{-8.314 \times 298.15}$

$+2 \Rightarrow K_p = e^{\frac{-141600}{-8.314 \times 298.15}}$

$= 6.44 \times 10^{24}$

(b) $\Delta G^\circ_{298} = -141.6 \text{ kJ} < 0$

$+2 \therefore$ this reaction is spontaneous

(c) $Q = \frac{P_{\text{SO}_3}^2}{P_{\text{SO}_2}^2 P_{\text{O}_2}} = \frac{0.1^2}{0.5^2 \cdot 0.01} = 4$

$Q < K_p$, the reaction goes right, the reaction process can't reach equilibrium.

(d) $\Delta G = \Delta G^\circ_{298} + \Delta G_f$

$+2 \Rightarrow = -141600 + 8.314 \times (273.15 + 25) \times \ln 4$

$= -141600 + 13436.372941$

$= -138163.63 \text{ J}$

$= -138.16 \text{ kJ/mol}$

$$(e) \Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -RT \ln K_p$$

$$\Rightarrow -198400 - (102 + 273.15) \times (-187.9) = -8.314 \times (102 + 273.15) \times \ln K_p$$

$$\Rightarrow -127909.315 = -8.314 \times 375.15 \times \ln K_p$$

$$\Rightarrow \ln K_p = 41.00995759$$

$$\Rightarrow K_p = 6.46 \times 10^{17}$$

$$(f) \Delta G_{298}^\circ = -141600 \text{ J}$$

$$+4. \Delta G_{298}^\circ = \Delta H^\circ - T\Delta S^\circ = -142405 \text{ J}$$

$$\therefore -141600 \approx -142405$$

$\therefore \Delta H^\circ$ and ΔS° do not change ^{much} with temperature.

$$\Delta G_{900}^\circ = -198400 - (900 + 273.15) \times (-187.9)$$

$$= 22.035 \text{ kJ}$$

$$\therefore \Delta G_{900}^\circ > 0$$

$\therefore$ nonspontaneous.

$$\begin{aligned} \Delta G_{\text{NaCl}}^{\circ} &= \Delta G_f^{\circ}(\text{Na}^+) + \Delta G_f^{\circ}(\text{Cl}^-) - \Delta G_f^{\circ}(\text{NaCl(s)}) \\ &= -261.9 \text{ kJ/mol} + (-131.2 \text{ kJ/mol}) - (-384.0 \text{ kJ/mol}) \\ &= -19.1 \text{ kJ/mol} \end{aligned}$$

+4.

$$\begin{aligned} \Delta G_{\text{AgCl}}^{\circ} &= \Delta G_f^{\circ}(\text{Ag}^+) + \Delta G_f^{\circ}(\text{Cl}^-) - \Delta G_f^{\circ}(\text{AgCl(s)}) \\ &= 77.1 \text{ kJ/mol} + (-131.2 \text{ kJ/mol}) - (-109.7 \text{ kJ/mol}) \\ &= -55.6 \text{ kJ/mol} \end{aligned}$$

$$\begin{aligned} \Delta G_{\text{NaCl}}^{\circ} &= \Delta H^{\circ} - T\Delta S^{\circ} \\ \Rightarrow -19100. &= 3600. - 298 \times \Delta S^{\circ} \\ \Rightarrow \Delta S_{\text{NaCl}}^{\circ} &= 42.62 \text{ J/K} \end{aligned}$$

+7

$$\begin{aligned} \Delta G_{\text{AgCl}}^{\circ} &= \Delta H^{\circ} - T\Delta S^{\circ} \\ \Rightarrow -55610 \text{ J/mol} &= 65700. - 298 \times \Delta S^{\circ} \end{aligned}$$

$$\Rightarrow \Delta S_{\text{AgCl}}^{\circ} = 33.86 \text{ J/K}$$

$$\therefore |\Delta S_{\text{NaCl}}^{\circ} - \Delta S_{\text{AgCl}}^{\circ}| < |\Delta H_{\text{NaCl}}^{\circ} - \Delta H_{\text{AgCl}}^{\circ}|$$

$\therefore$ enthalpy term is the primary reason.

(C) NaCl:
 $\Delta G^\circ = -9100 \text{ J/mol} = -RT \ln K_{sp}$

$$\Rightarrow \ln K_{sp} = \frac{-9100}{-8.314 \times 298} \approx 3.693$$

$$\Rightarrow K_{sp} \approx 39.37$$

AgCl:

$$\Delta G^\circ = 55610 \text{ J/mol} = -RT \ln K_{sp}$$

+4. $\Rightarrow \ln K_{sp} = \frac{55610}{-8.314 \times 298} \approx -22.45$

$$\Rightarrow K_{sp} \approx 1.78 \times 10^{-10}$$

(d). $\Delta G = \Delta H - T\Delta S$

ΔH and ΔS do not change with temperature

+3.

$$\Rightarrow \Delta G = -T\Delta S + \Delta H \rightarrow \text{constant}$$

$\Rightarrow \Delta G$ goes down with the increasing temperature.

$\therefore$ the solution process goes right more.